THQ #7

By: Ethan Le :D

Original Problem

Objective Function:

$$\textbf{max } z = 11x_1 + 4x_2 + x_3 + 15x_4$$

Constraints:

- $3x_1 + x_2 + 2x_3 + 4x_4 \leq 28$
- $8x_1 + 2x_2 - x_3 + 7x_4 \leq 50$
- $x_1, x_2, x_3, x_4 \geq 0$ **(non-negative)**

Given Tableaux (initial and final):

THQ# 7 – Chapter 5-Part 2- MATH177- Yahdi.

The Take Home Quiz is to submit via canvas by the indicated due date by 11:59 pm. Please have all submitted pages together as one PDF file using any appropriate method (THQ information).

Original Problem: Consider the following problem with the following initial and final tableaux.

Maximizing $z = 11x_1 + 4x_2 + x_3 + 15x_4$
subject to
$3x_1 + x_2 + 2x_3 + 4x_4 \leq 28$
$8x_1 + 2x_2 - x_3 + 7x_4 \leq 50$
$x_1, x_2, x_3, x_4 \geq 0$

	x_1	x_2	x_3	x_4	x_5	x_6	
x_5	3	1	2	4	1	0	28
x_6	8	2	−1	7	0	1	50
	−11	−4	−1	−15	0	0	0
x_4	−2	0	5	1	2	−1	6
x_2	11	1	−18	0	−7	4	4
	3	0	2	0	2	1	106

Section 5.4:

Suppose a variable x_7 is added to the **Original Problem** such that the problem becomes:

$$Max\ 11x_1 + 4x_2 + x_3 + 15x_4 + \alpha x_7$$

- $3x_1 + x_2 + 2x_3 + 4x_4 + \beta x_7 \leq 28$
- $8x_1 + 2x_2 - x_3 + 7x_4 + \gamma x_7 \leq 50$
- $x_1, x_2, x_3, x_4, x_7 \geq 0$

Solve the problem for each of the following cases using the given original tableaux (without x_7) above.

1. $\alpha = 9, \beta = 2, \gamma = 5$
2. $\alpha = 27, \beta = 7, \gamma = 12$

Figure 1: Initial and final tableaux

(In the tableaux, x_5 and x_6 are the slack variables for the two constraints.)

Initial basis (ordered to form the identity in A):

$$B_0 = [x_5,\ x_6]$$

Final basis (ordered to form the identity in A^*):

$$B = [x_4,\ x_2]$$

$$A_B^{-1} = A_{B_0}^* = \text{final-tableau columns under } x_5, x_6 :$$

$$A_B^{-1} = A_{B_0}^* = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$$

Question 1 (Section 5.4)

Suppose a new variable x_7 is added to the original problem such that the problem becomes:

Objective Function:

$$\textbf{max } 11x_1 + 4x_2 + x_3 + 15x_4 + \alpha x_7$$

Constraints:

- $3x_1 + x_2 + 2x_3 + 4x_4 + \beta x_7 \leq 28$
- $8x_1 + 2x_2 - x_3 + 7x_4 + \gamma x_7 \leq 50$
- $x_1, x_2, x_3, x_4, x_7 \geq 0$ **(non-negative)**

Solve the problem for each of the following cases using the given original tableaux (without x_7) above.

Tasks:

Using the handout (Section 5.4) with the **tableau / minimize** c-row (initial bottom row is $-$max coeffs):

- $c_B^\top = [-15,\ -4]$ for $B = [x_4,\ x_2]$
- $c_{n+1} = -\alpha$
- If $c_{n+1}^* \geq 0 \rightarrow x_7$ stays out, current BFS remains optimal
- If $c_{n+1}^* < 0 \rightarrow$ enter x_7 and pivot

1. $\alpha = 9$, $\beta = 2$, $\gamma = 5$

$$A_{n+1} = \begin{bmatrix} \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$A_{n+1}^* = A_B^{-1} A_{n+1} = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 4-5 \\ -14+20 \end{bmatrix} = \begin{bmatrix} -1 \\ 6 \end{bmatrix}$$

$$c_{n+1} = -\alpha = -9, \qquad c_B^\top = [-15,\ -4]$$

$$c_{n+1}^* = c_{n+1} - c_B^\top A_{n+1}^* = -9 - [-15,\ -4] \begin{bmatrix} -1 \\ 6 \end{bmatrix} = -9 - (15-24) = -9 - (-9) = 0$$

Since $c_{n+1}^* = 0 \geq 0$, x_7 is not a basic variable ($x_7 = 0$) and the current BFS remains optimal:

$$(x_1, x_2, x_3, x_4, x_7) = (0,\ 4,\ 0,\ 6,\ 0), \qquad z^* = 106.$$

2. $\alpha = 27$, $\beta = 7$, $\gamma = 12$

$$A_{n+1} = \begin{bmatrix} \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 7 \\ 12 \end{bmatrix}$$

$$A^*_{n+1} = A_B^{-1} A_{n+1} = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix} \begin{bmatrix} 7 \\ 12 \end{bmatrix} = \begin{bmatrix} 14 - 12 \\ -49 + 48 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$c_{n+1} = -\alpha = -27, \qquad c_B^\top = [-15,\ -4]$$

$$c^*_{n+1} = c_{n+1} - c_B^\top A^*_{n+1} = -27 - [-15,\ -4] \begin{bmatrix} 2 \\ -1 \end{bmatrix} = -27 - (-30 + 4) = -27 - (-26) = -1$$

Since $c^*_{n+1} = -1 < 0$, enter x_7. Append its column to the final tableau (slack columns stay labeled x_5, x_6 as in the given tableaux):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_4	-2	0	5	1	2	-1	2	6
x_2	11	1	-18	0	-7	4	-1	4
z	3	0	2	0	2	1	-1	106

Ratio test on the entering column $[2,\ -1]^\top$:

- x_4: $6/2 = 3$
- x_2: $-1 < 0$, cannot leave

Pivot on the 2: x_7 enters, x_4 leaves.

After the pivot (divide the x_4 row by 2, then add that row to the x_2 and z rows):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_7	-1	0	$\frac{5}{2}$	$\frac{1}{2}$	1	$-\frac{1}{2}$	1	3
x_2	10	1	$-\frac{31}{2}$	$\frac{1}{2}$	-6	$\frac{7}{2}$	0	7
z	2	0	$\frac{9}{2}$	$\frac{1}{2}$	3	$\frac{1}{2}$	0	109

All z-row entries are ≥ 0, so this is optimal:

$$(x_1, x_2, x_3, x_4, x_7) = (0,\ 7,\ 0,\ 0,\ 3), \qquad z^* = 109.$$

Check: $4(7) + 27(3) = 28 + 81 = 109$.

Question 2 (Section 5.6)

Suppose changes were made to the **Original Problem** (without x_7) such that 50 in the RHS of the second constraint is changed to $50 + \lambda$.

Use the given original initial/final tableaux (and $A_B^{-1} = A^*_{B_0}$ from above).

Tasks:

1. What is the new expression of b^*_λ and z^*_λ?

$$b_\lambda = \begin{bmatrix} 28 \\ 50+\lambda \end{bmatrix}$$

$$b^*_\lambda = A_B^{-1} b_\lambda = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix} \begin{bmatrix} 28 \\ 50+\lambda \end{bmatrix} = \begin{bmatrix} 56-(50+\lambda) \\ -196+4(50+\lambda) \end{bmatrix} = \begin{bmatrix} 6-\lambda \\ 4+4\lambda \end{bmatrix}$$

(Check at $\lambda = 0$: $b^* = [6,\ 4]^\top$, matching the final tableau.)

For z^*_λ, use the max-form costs of $B = [x_4,\ x_2]$ (handout: opposite sign if working from the min/tableau c-row):

$$c_B^\top = [15,\ 4], \qquad z_0 = 0$$

$$z^*_\lambda = z_0 + c_B^\top b^*_\lambda = [15,\ 4] \begin{bmatrix} 6-\lambda \\ 4+4\lambda \end{bmatrix} = 15(6-\lambda) + 4(4+4\lambda) = 90 - 15\lambda + 16 + 16\lambda = 106 + \lambda.$$

Same result from the dual (shadow prices from the final z-row under x_5, x_6: $y = [2,\ 1]$):

$$z^*_\lambda = 28(2) + (50+\lambda)(1) = 56 + 50 + \lambda = 106 + \lambda.$$

2. Starting from the given tableaux above and with any additional tableaux if needed, solve for the optimal value and point if $\lambda = -2$.

Plug in $\lambda = -2$:

$$b^*_{-2} = \begin{bmatrix} 6-(-2) \\ 4+4(-2) \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \end{bmatrix}, \qquad 106 + (-2) = 104.$$

The second basic value is negative, so the current basis is **not feasible**. Use the **dual simplex** algorithm on the final tableau with the updated RHS:

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_4	-2	0	5	1	2	-1	8
x_2	11	1	-18	0	-7	4	-4
z	3	0	2	0	2	1	104

Exit: most negative RHS $\to$ x_2 row (RHS -4).

Enter: among $a_{rj} < 0$ in that row, choose the smallest $|c_j/a_{rj}|$:

- x_3: $a = -18$, ratio $2/(-18) = -1/9$, $|\text{ratio}| = 1/9$
- x_5: $a = -7$, ratio $2/(-7) = -2/7$, $|\text{ratio}| = 2/7$

Smallest absolute ratio is $1/9 \to$ enter x_3. Pivot on -18.

After the pivot (divide the x_2 row by -18, then clear the x_3 column in the other rows):

	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_4	$\frac{19}{18}$	$\frac{5}{18}$	0	1	$\frac{1}{18}$	$\frac{1}{9}$	$\frac{62}{9}$
x_3	$-\frac{11}{18}$	$-\frac{1}{18}$	1	0	$\frac{7}{18}$	$-\frac{2}{9}$	$\frac{2}{9}$
z	$\frac{38}{9}$	$\frac{1}{9}$	0	0	$\frac{11}{9}$	$\frac{13}{9}$	$\frac{932}{9}$

All RHS entries are ≥ 0 and all z-row entries are ≥ 0, so this is optimal:

$$(x_1, x_2, x_3, x_4) = \left(0,\ 0,\ \frac{2}{9},\ \frac{62}{9}\right), \qquad z^* = \frac{932}{9}.$$

Check: $1 \cdot \frac{2}{9} + 15 \cdot \frac{62}{9} = \frac{2 + 930}{9} = \frac{932}{9}$.

Question 3 (Section 5.7)

Going back to the **Original Problem** (without x_7 and with 50 in the RHS), for each case (separately) below, solve after adding the given constraint.

Original optimal point (before the new constraint): $(x_1, x_2, x_3, x_4) = (0, 4, 0, 6)$, $z^* = 106$.

From the final tableau:

$$x_4 = 6 + 2x_1 - 5x_3 - 2x_5 + x_6, \qquad x_2 = 4 - 11x_1 + 18x_3 + 7x_5 - 4x_6.$$

1. Add $3x_1 + x_2 + 3x_4 \leq 24$

Check the current optimum: $3(0) + 4 + 3(6) = 22 \leq 24$. The new constraint is already satisfied, so the BFS stays optimal:

$$(x_1, x_2, x_3, x_4) = (0,\ 4,\ 0,\ 6), \qquad z^* = 106.$$

(With slack $s = 24 - 22 = 2$ for the new constraint.)

2. Add $3x_1 + x_2 + 3x_4 \leq 20$

Check: $22 \leq 20$ is false, so the current point is infeasible. Add slack x_7: $3x_1 + x_2 + 3x_4 + x_7 = 20$.

First append the **original** new constraint as a row in the final tableau (zeros under the old slack columns x_5, x_6 for now):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_4	-2	0	5	1	2	-1	0	6
x_2	11	1	-18	0	-7	4	0	4
x_7	3	1	0	3	0	0	1	20
z	3	0	2	0	2	1	0	106

This x_7 row still has nonzero entries under the current basics x_2 and x_4. Clear those by row operations:

$$(\text{new } x_7 \text{ row}) \ - \ 1 \cdot (x_2 \text{ row}) \ - \ 3 \cdot (x_4 \text{ row}) :$$

$$
\begin{aligned}
&[3,\ 1,\ 0,\ 3,\ 0,\ 0,\ 1 \mid 20] \\
&-\ [11,\ 1,\ -18,\ 0,\ -7,\ 4,\ 0 \mid 4] \\
&-\ 3[-2,\ 0,\ 5,\ 1,\ 2,\ -1,\ 0 \mid 6] \\
&=\ [-2,\ 0,\ 3,\ 0,\ 1,\ -1,\ 1 \mid -2].
\end{aligned}
$$

Updated tableau after rewriting the new constraint:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_4	-2	0	5	1	2	-1	0	6
x_2	11	1	-18	0	-7	4	0	4
x_7	-2	0	3	0	1	-1	1	-2
z	3	0	2	0	2	1	0	106

(The RHS -2 is $20 - 22$: the old optimum overshoots the new limit by 2.)

RHS of x_7 is negative $\rightarrow$ **dual simplex**.

Exit x_7. Negative entries in that row: x_1 (-2), x_6 (-1).

Ratios $|c_j/a_{rj}|$: x_1: $|3/(-2)| = 3/2$; x_6: $|1/(-1)| = 1$.

Enter x_6 (smaller absolute ratio). Pivot on -1:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_4	0	0	2	1	1	0	-1	8
x_2	3	1	-6	0	-3	0	4	-4
x_6	2	0	-3	0	-1	1	-1	2
z	1	0	5	0	3	0	1	104

Still have a negative RHS $\rightarrow$ dual simplex again.

Exit x_2. Negative entries: x_3 (-6), x_5 (-3).

Ratios: x_3: $|5/(-6)| = 5/6$; x_5: $|3/(-3)| = 1$.

Enter x_3. Pivot on -6:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
x_4	1	$\frac{1}{3}$	0	1	0	0	$\frac{1}{3}$	$\frac{20}{3}$
x_3	$-\frac{1}{2}$	$-\frac{1}{6}$	1	0	$\frac{1}{2}$	0	$-\frac{2}{3}$	$\frac{2}{3}$
x_6	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	$\frac{1}{2}$	1	-3	4
z	$\frac{7}{2}$	$\frac{5}{6}$	0	0	$\frac{1}{2}$	0	$\frac{13}{3}$	$\frac{302}{3}$

All RHS ≥ 0 and all z-row entries ≥ 0, so optimal:

$$
(x_1, x_2, x_3, x_4) = \left(0,\ 0,\ \frac{2}{3},\ \frac{20}{3}\right), \qquad z^* = \frac{302}{3}.
$$

Checks: - New constraint: $3 \cdot \dfrac{20}{3} = 20 \leq 20$ - Objective: $1 \cdot \dfrac{2}{3} + 15 \cdot \dfrac{20}{3} = \dfrac{302}{3}$